15.9 Solution: (a) For a screened Coulomb potential,

$$V(r) = \frac{zZe^2}{r} e^{-\alpha r},$$

the force on the particle is

$$\mathbf{F} = -\nabla V = zZe^2 \left(\alpha + \frac{1}{r}\right) \frac{e^{-\alpha r}}{r} \hat{r}.$$

In its componentwise form, by the Newton's second law, we have

$$m\ddot{x}(t) = zZe^2 \left(\alpha + \frac{1}{r}\right) \frac{e^{-\alpha r}}{r} \frac{x}{r}, \quad m\ddot{y}(t) = zZe^2 \left(\alpha + \frac{1}{r}\right) \frac{e^{-\alpha r}}{r} \frac{b}{r}.$$

Here, we have assumed that the particle follows a straight line, so that $y \equiv b$ and $r = \sqrt{x^2 + b^2}$. To apply the dipole approximation (see Problem 14.22), we need the Fourier transform of the accelerations,

$$\begin{aligned}
\ddot{x}(\omega) &= \int_{-\infty}^{+\infty} \ddot{x}(t) e^{i\omega t} dt = \frac{zZe^2}{m} \int_{-\infty}^{+\infty} \left(\alpha + \frac{1}{r}\right) \frac{e^{-\alpha r}}{r} \frac{x}{r} e^{i\omega t} dt \\
&= \frac{\alpha zZe^2}{m} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha \left(x^2 + b^2\right)^{3/2}}\right) x \exp\left(-\alpha\sqrt{x^2 + b^2}\right) e^{i\omega t} dt.
\end{aligned}$$

Since $x = vt$ for a straight path, and also notice the parity of the integrand, we can write the Fourier transform of the acceleration in the x-direction as

$$\ddot{x}(\omega) = \frac{i\alpha zZe^2}{mv} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha \left(x^2 + b^2\right)^{3/2}}\right) x \exp\left(-\alpha\sqrt{x^2 + b^2}\right) \sin\left(\frac{\omega}{v} x\right) dx.$$

Similarly,

$$\ddot{y}(\omega) = \frac{\alpha b zZe^2}{mv} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha \left(x^2 + b^2\right)^{3/2}}\right) \exp\left(-\alpha\sqrt{x^2 + b^2}\right) \cos\left(\frac{\omega}{v} x\right) dx.$$

Using the integration identities (see Gradshteyn and Ryzhik, 8^{th} ed., 3.914.5 and 3.914.10, pg. 522),

$$\int_0^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha \left(x^2 + b^2\right)^{3/2}}\right) x \exp\left(-\alpha\sqrt{x^2 + b^2}\right) \sin\left(\gamma x\right) dx = \frac{\gamma}{\alpha} K_0(b\sqrt{\alpha^2 + \gamma^2}),$$

$$\int_0^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha \left(x^2 + b^2\right)^{3/2}}\right) \exp\left(-\alpha\sqrt{x^2 + b^2}\right) \cos\left(\frac{\omega}{v} x\right) dx = \frac{\sqrt{\alpha^2 + \gamma^2}}{\alpha b} K_1(b\sqrt{\alpha^2 + \gamma^2}),$$

the above Fourier transforms can be *exactly* evaluated, with $\gamma \to \omega/v$,

$$\ddot{x}(\omega) = \frac{2i\omega zZe^2}{mv^2} K_0\left(b\sqrt{\alpha^2 + \frac{\omega^2}{v^2}}\right), \quad \ddot{y}(\omega) = \frac{2zZe^2}{mv} \sqrt{\alpha^2 + \frac{\omega^2}{v^2}} K_1\left(b\sqrt{\alpha^2 + \frac{\omega^2}{v^2}}\right).$$

From the dipole approximation,

$$\frac{dI}{d\omega}(\omega, b) = \frac{2}{3\pi c^3} \left|\ddot{\mathbf{d}}(\omega)\right|^2,$$

where $\ddot{\mathbf{d}}(\omega) = ze(\ddot{x}(\omega), \ddot{y}(\omega))$. Thus,

$$\frac{dI}{d\omega}(\omega, b) = \frac{8}{3\pi c^3}\left(\frac{Zz^2e^3}{mv}\right)^2\left[\frac{\omega^2}{v^2}K_0\left(b\sqrt{\alpha^2+\frac{\omega^2}{v^2}}\right)^2 + \left(\alpha^2+\frac{\omega^2}{v^2}\right)K_1\left(b\sqrt{\alpha^2+\frac{\omega^2}{v^2}}\right)^2\right].$$

For $\omega v/b \gg 1$, $b\sqrt{\alpha^2+\omega^2/v^2} \gg 1$. By the asymptotic behavior of the modified Bessel function, $K_\nu(x) \sim e^{-x}$ as $x \to +\infty$, we can see that the radiated energy is almost negligible. On the other, when $\omega \ll v/b$, we can safely drop ω/v in the above expression and will obtain

$$\frac{dI}{d\omega}(\omega, b) = \frac{8}{3\pi c^3}\left(\frac{Zz^2e^3}{mv}\right)^2\alpha^2K_1(\alpha b)^2 = \frac{8}{3\pi}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\alpha^2K_1(\alpha b)^2.$$

(b) The radiation cross section is

$$\begin{aligned}\frac{d\chi}{d\omega} &= \int_{b_{\min}}^{b_{\max}}\frac{dI}{d\omega}(\omega, b)\cdot 2\pi b db = \frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\int_{b_{\min}}^{b_{\max}}\alpha^2K_1(\alpha b)^2 b db \\ &= \frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\int_{\alpha b_{\min}}^{\alpha b_{\max}} xK_1(x)^2dx.\end{aligned}$$

Since

$$\begin{aligned}\int xK_1(x)^2dx &= \frac{1}{2}x^2\left[K_1(x)^2 - K_0(x)K_2(x)\right] + C \\ &= \frac{1}{2}x^2\left[K_1(x)^2 - K_0(x)\left(K_0(x)+\frac{2}{x}K_1(x)\right)\right] + C \\ &= -\frac{1}{2}x^2\left[K_0(x)^2 - K_1(x)^2 + \frac{2}{x}K_0(x)K_1(x)\right] + C,\end{aligned}$$

then

$$\frac{d\chi}{d\omega} = -\frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\left\{\frac{1}{2}x^2\left[K_0(x)^2 - K_1(x)^2 + \frac{2}{x}K_0(x)K_1(x)\right]\right\}\Bigg|_{\alpha b_{\min}}^{\alpha b_{\max}}.$$